EEE 120

Lab 3 Answer Sheet (Online Class)

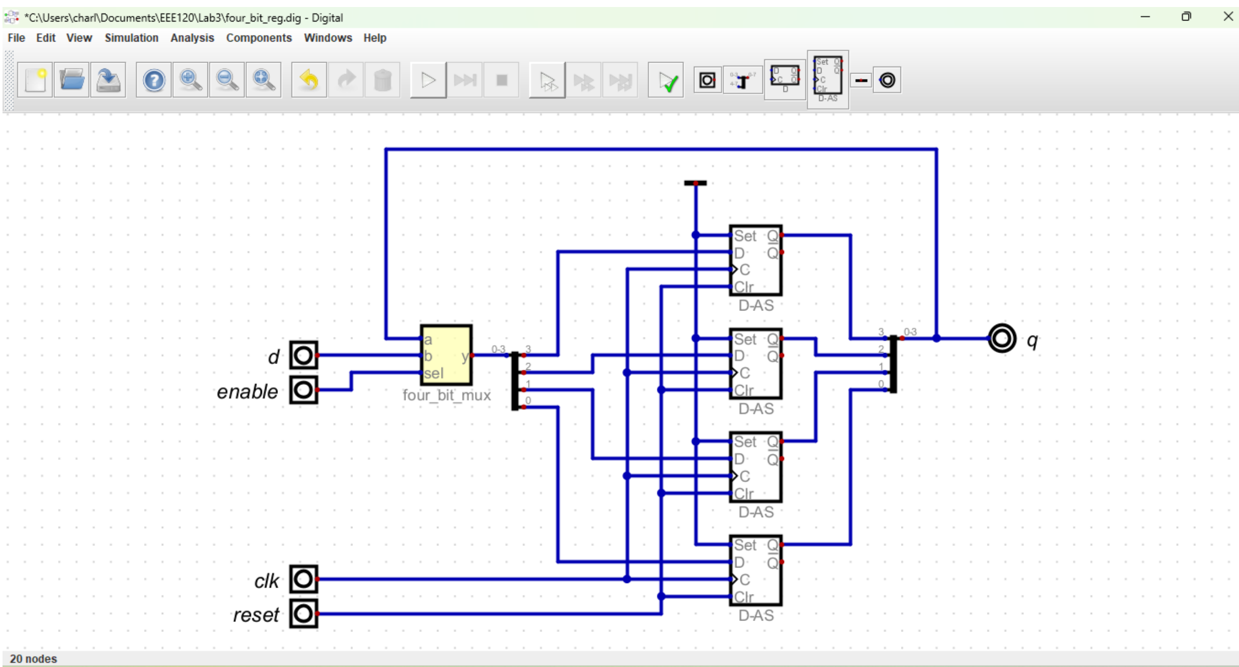
Registers, Counters and the “Brainless CPU”

Name:____Charlene Butler____

Semester/Year/Session (A/B):___Summer 2024 C_ Date:_06/16/2024_______________

Task 3-1: Build and Test a 4-Bit D Register with Enable

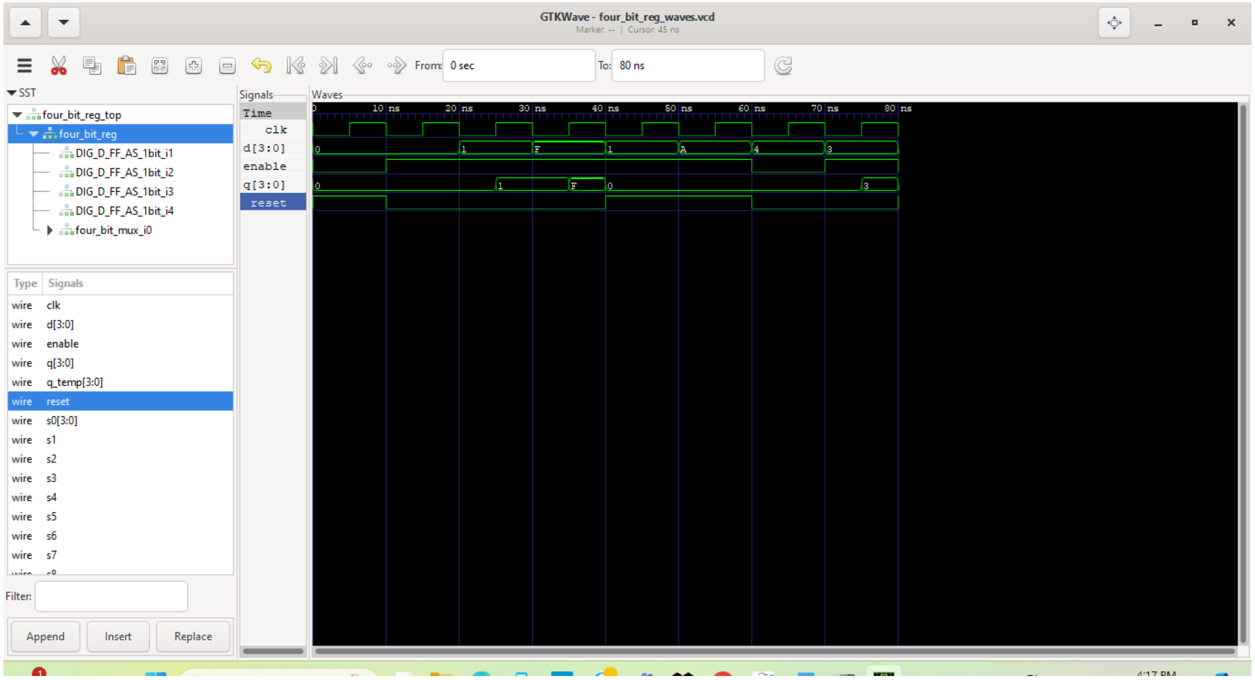
Include a picture of your Digital circuit here:



Please comment on the single biggest issue you were facing when designing the circuit.

My biggest issue was realizing how it all worked.

Include a picture of your GTKWave waveforms (timing diagram) here:



Did the circuit behave as expected? If no, what was wrong?

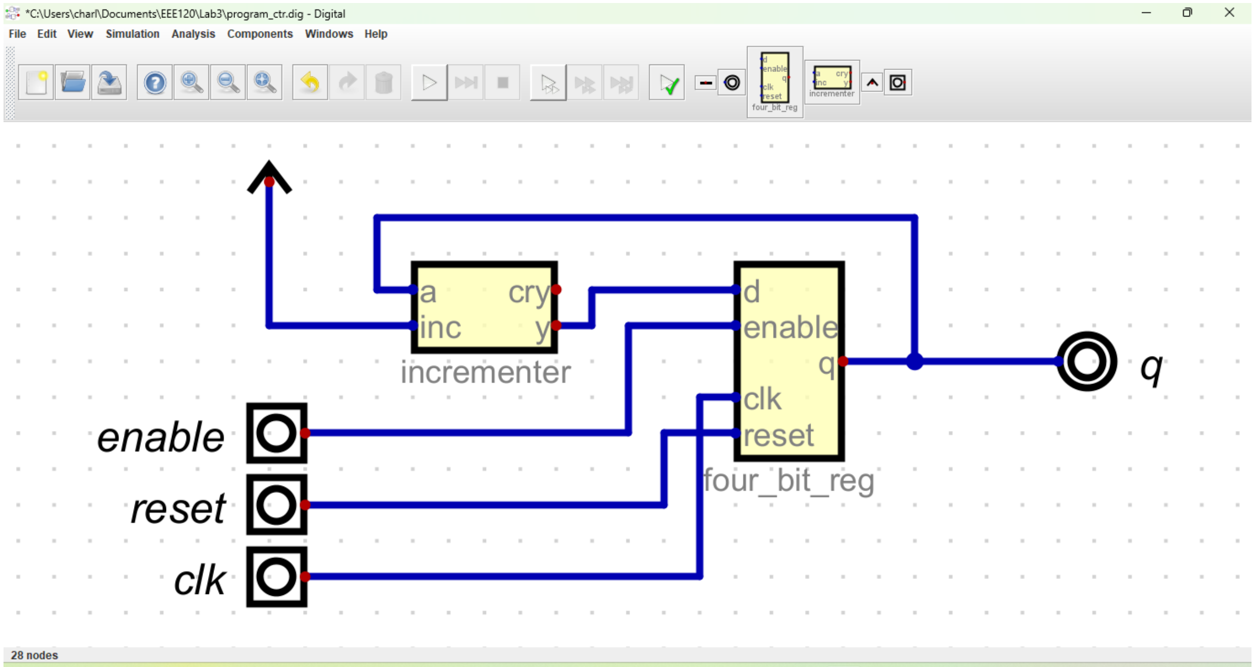
It behaved exactly like I expected it to.

Please comment on the single biggest issue you were facing when simulating the circuit.

No issues on this one at all.

Task 3-2: Build and Test a 4-Bit UP Counter

Include a picture of your Digital circuit here:



Please comment on the single biggest issue you were facing when designing the circuit.

No issue with this one.

Did the circuit behave as expected? If no, what was wrong?

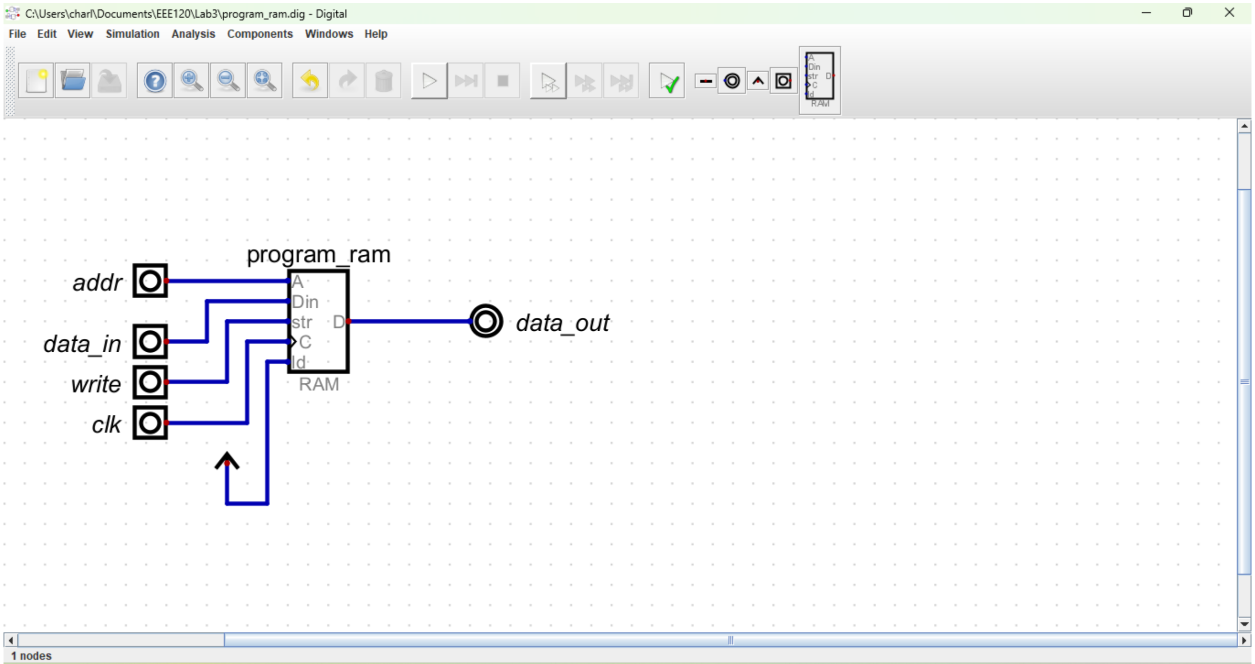
It behaved exactly as expected.

Please comment on the single biggest issue you were facing when simulating the circuit.

No issue.

Task 3-3: Create a 4-Bit RAM with 16 4-Bit Words

Include a picture of your Digital circuit here:



Please comment on the single biggest issue you were facing when designing the circuit.

No issues.

Did the circuit behave as expected? If no, what was wrong?

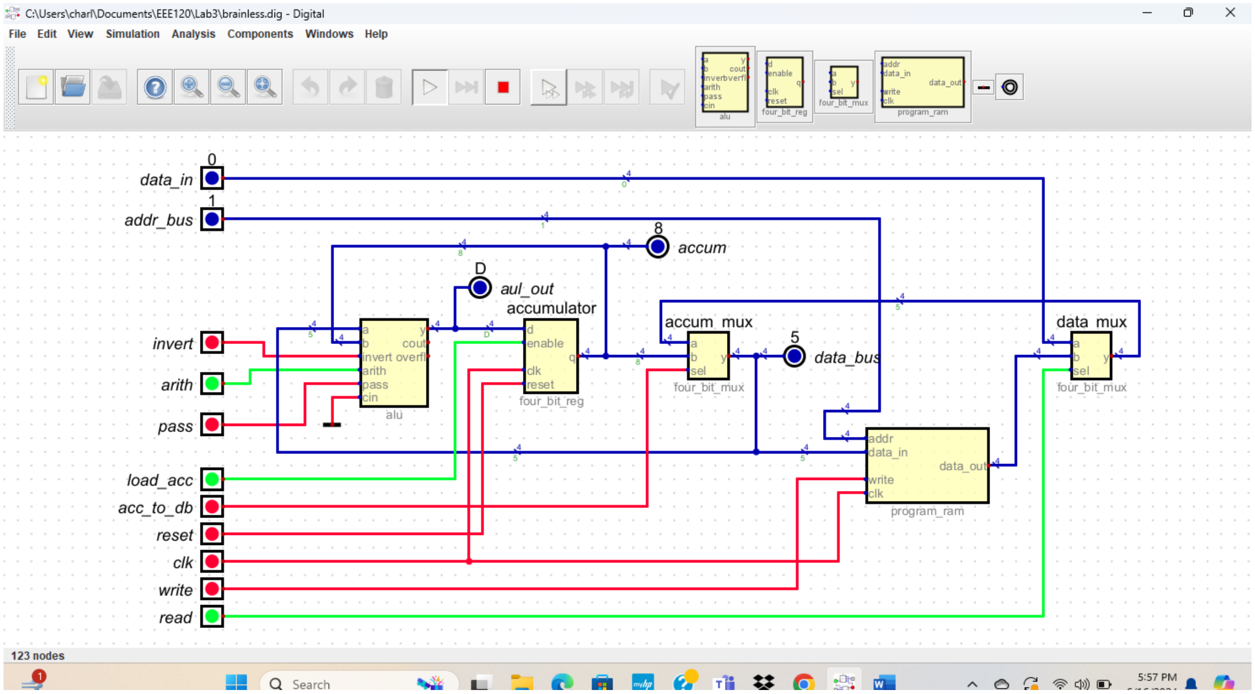
Exactly as expected.

Please comment on the single biggest issue you were facing when simulating the circuit.

No issues.

Task 3-4: Build and Test the Brainless Central Processing Unit

Include a picture of your Digital circuit here:



Please comment on the single biggest issue you were facing when designing the circuit.

No issues.

Did the circuit behave as expected? If no, what was wrong?

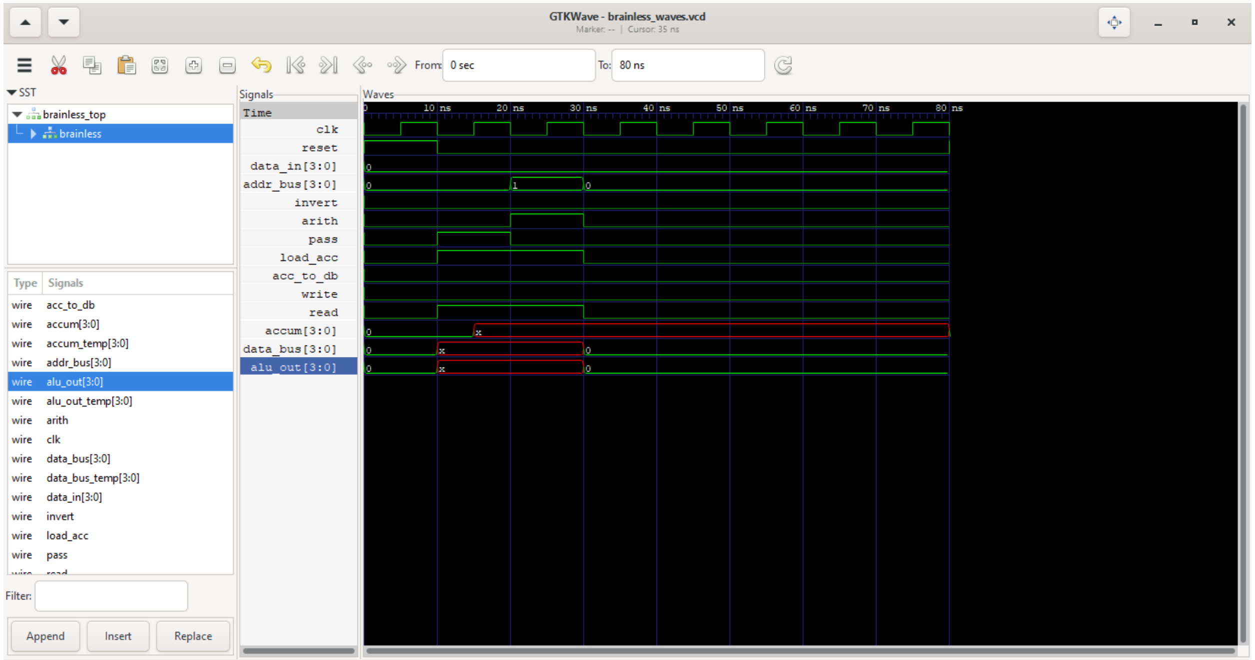
Yes, it behaved as expected.

Please comment on the single biggest issue you were facing when simulating the circuit.

No issues.

Task 3-5: Simulate the Brainless Central Processing Unit

Include a picture of your GTKWave waveforms (timing diagram) here:



Did the circuit behave as expected? If no, what was wrong?

Yes, the digital circuit works just fine, but I can not figure out what happened with my waves. They should have 3, 5, and 8 and NOT x.

Please comment on the single biggest issue you were facing when simulating the circuit.

See the answer above.

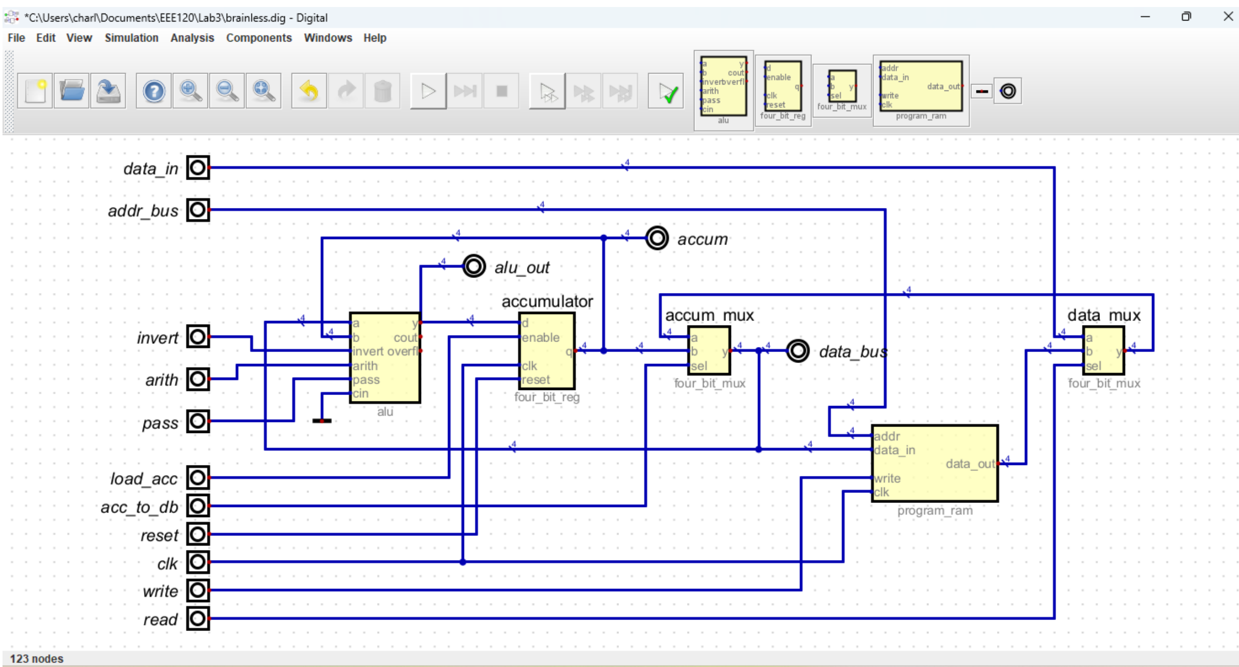
Task 3-6: Create Additional Tests

In order to stay organized, you may use the table here to map out the different values for each test. Repeat this table for each of the additional ALU operations.

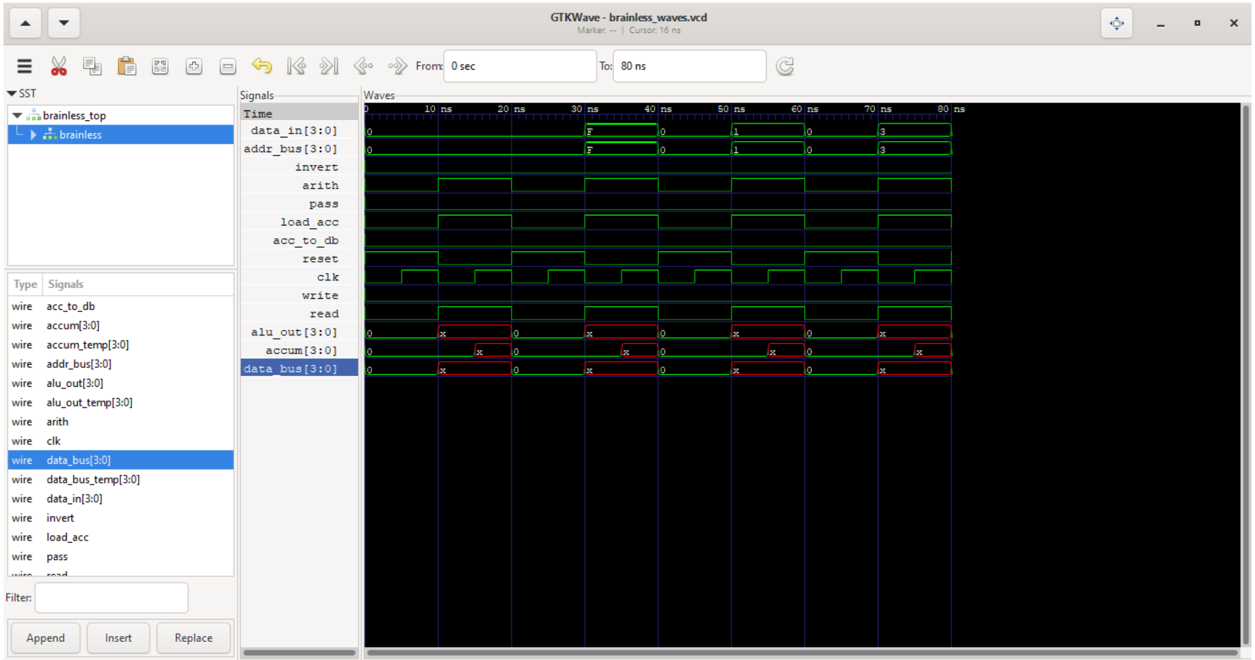
Table 1	
Operation [Add operand to Accumulator (ACC)]	
Control Line	Value
data_in	0
addr_bus	Address of operand
write	0
read	1
acc_to_db	0

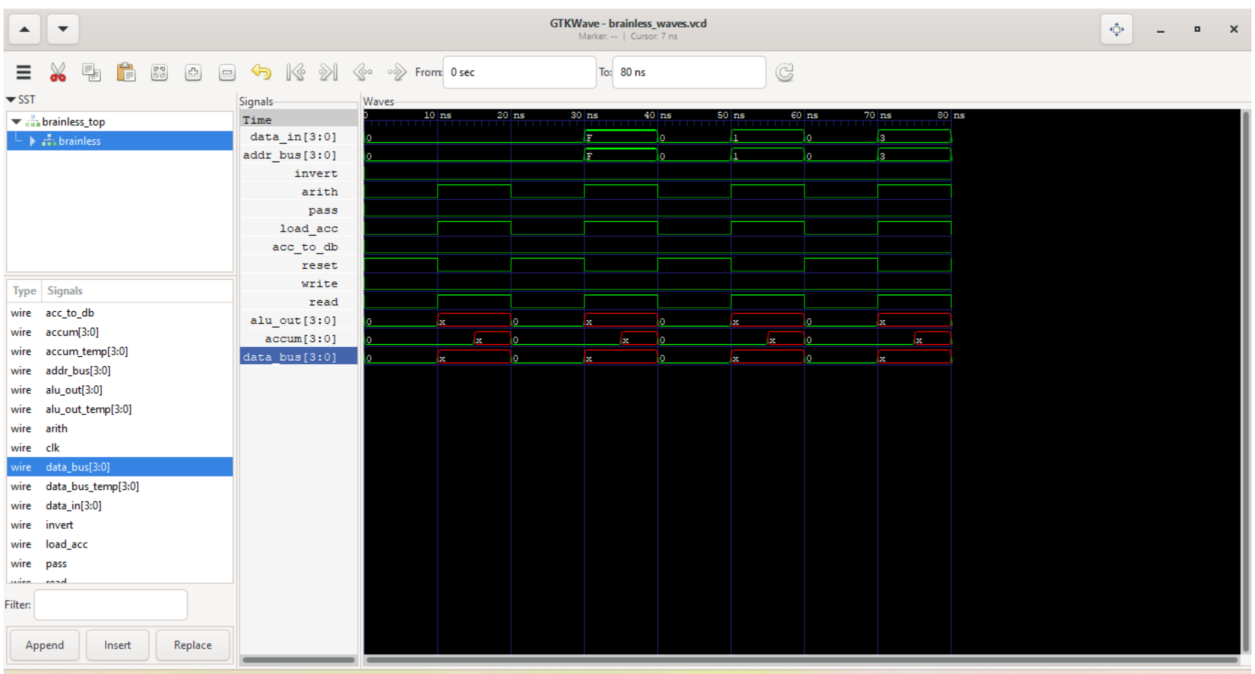
load_acc	1
pass	0
invert	0
arith	1

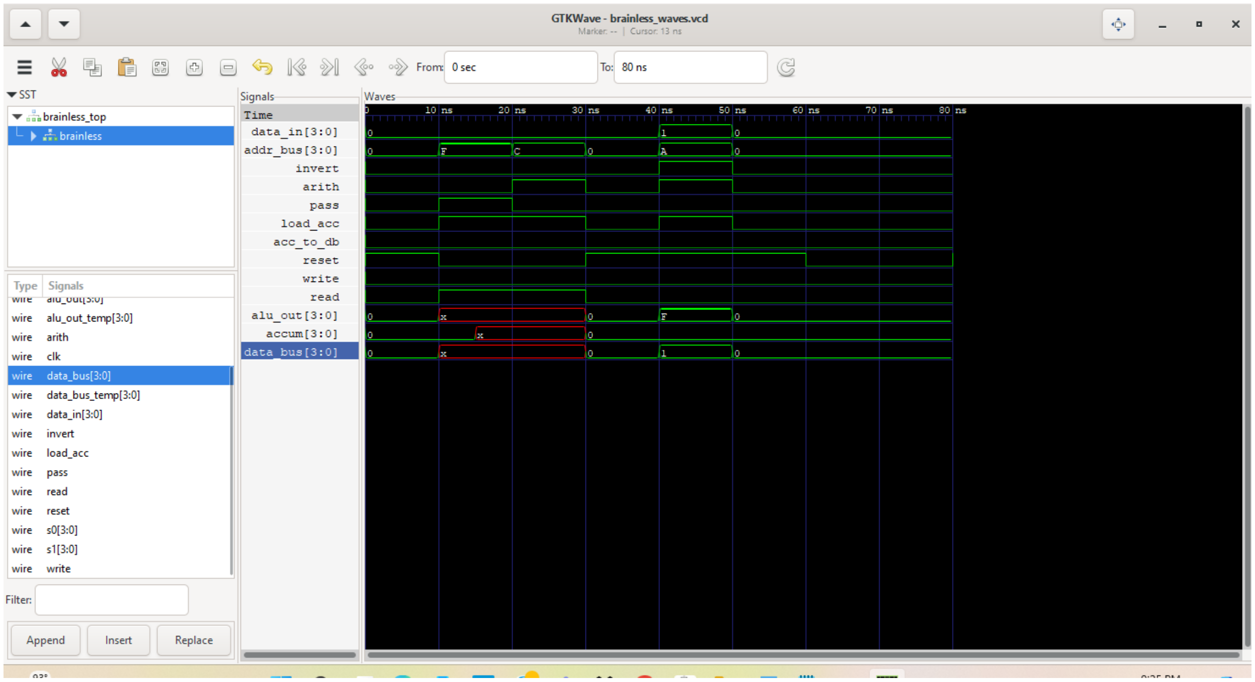
Include a picture of your Digital circuit here:



Include a picture of your GTKWave waveforms here (one per required test):


GTKWave - brainless_waves.vcd
Marker: -- | Cursor: 16 ns
From: 0 sec
To: 80 ns
SST
brainless_top
brainless
Type
Signals
wire acc_to_db
wire accum[3:0]
wire accum_temp[3:0]
wire addr_bus[3:0]
wire alu_out[3:0]
wire alu_out_temp[3:0]
wire arith
wire clk
wire data_bus[3:0]
wire data_bus_temp[3:0]
wire data_in[3:0]
wire invert
wire load_acc
wire pass
Filter:
Append
Insert
Replace
Signals
Time
data_in[3:0]
addr_bus[3:0]
invert
arith
pass
load_acc
acc_to_db
reset
clk
write
read
alu_out[3:0]
accum[3:0]
data_bus[3:0]
Waves
10 ns
20 ns
30 ns
40 ns
50 ns
60 ns
70 ns
80 ns


GTKWave - brainless_waves.vcd
Marker: -- | Cursor: 7 ns
From: 0 sec
To: 80 ns
SST
brainless_top
brainless
Type
Signals
wire acc_to_db
wire accum[3:0]
wire accum_temp[3:0]
wire addr_bus[3:0]
wire alu_out[3:0]
wire alu_out_temp[3:0]
wire arith
wire clk
wire data_bus[3:0]
wire data_bus_temp[3:0]
wire data_in[3:0]
wire invert
wire load_acc
wire pass
Filter:
Append
Insert
Replace
Signals
Time
data_in[3:0]
addr_bus[3:0]
invert
arith
pass
load_acc
acc_to_db
reset
write
read
alu_out[3:0]
accum[3:0]
data_bus[3:0]
Waves
10 ns
20 ns
30 ns
40 ns
50 ns
60 ns
70 ns
80 ns



Please comment on the single biggest issue you were facing when designing the circuit.

No issues.

Did the circuit behave as expected? If no, what was wrong?

No, it is showing x's in the waveform. I have tried everything to get this to work properly.

Please comment on the single biggest issue you were facing when simulating the circuit.

The waveform.

Task 3-7: Create a video and submit your report (Optional)

[This task is useful to get partial credit if your schematic is not working. Take advantage of it to explain to the grader your understanding of the circuit. More importantly, explain where you think the mistake is in and what you would do if you were given more time to fix it.]

Record a short video showing your schematic in Digital and your waveforms in GTKWave. Be sure to show yourself in the video and show your screen. **Upload the video to your Google Drive (personal one or ASU one). Copy and paste the sharing link to that video here. Make sure the link is working and pointing to the correct video. Do NOT upload your video to YouTube.** If your circuit is not working as expected, explain in the video how it is not working and where you expect the mistake to be from.

Video Link:

At the beginning of your recording, say your name, the task number and circuit name. Be brief in your recording. Submit the completed template to Canvas.

Make sure all your files are in the Lab3 directory. Create a zip file of the Lab3 directory. Remember to turn in the zip file and your completed template on Canvas! Make sure you upload the template before the zip file.

LAB 3: LAB REPORT GRADE SHEET

Name ________________________________

Instructor Assessment

Grading Criteria	Max Points	Points Lost
Description of Assigned Tasks, Work Performed & Outcomes Met		
Task 3-1: Build and Test a 4-Bit D Register with Enable	10	
Task 3-2: Build and Test a 4-Bit UP Counter	10	
Task 3-3: Create a 4-Bit RAM with 16 4-Bit Words	10	
Task 3-4: Build and Test the Brainless Central Processing Unit	10	
Task 3-5: Simulate the Brainless Central Processing Unit	10	
Task 3-6: Create Additional Tests	30	
Task 3-7: Create a video and submit your report (Optional)		
Lab Score (80 points total)	**Points Lost**	
	Late Lab	
	Lab Score	